

Exam: Discrete Probability Distributions

Practice Exam with Solutions | Basic Level

1. **[Application] (4 Pts.)** MC test: 10 questions, 4 options each, guessing. Distribution? $P(X=4)$? $P(X \geq 7)$?
2. **[Application] (4 Pts.)** Hospital averages 3.2 emergencies per hour. Poisson. a) $P(X=0)$ in 1h? b) $P(X>5)$ in 2h?
3. **[Open question] (3 Pts.)** Lot of 30 devices, 4 defective. Sample 5 w/o replacement. $P(\text{exactly 1 defective})$?
4. **[Open question] (3 Pts.)** Die rolled until first 6. Distribution? $P(\text{first 6 on 3rd roll})$? Expected rolls?
5. **[Open question] (4 Pts.)** $B(200, 0.04)$ approximated by Poisson. Why? $P(X \leq 3)$? Compare expectations.

SOLUTIONS

1. Binomial $B(10, 0.25)$. $P(X=4)=0.1460$. $P(X \geq 7)=0.00352$.

2. a) $e^{-3.2}=0.0408$. b) $\lambda=6.4$, $P(X>5)=0.617$.

3. $H(30, 4, 5)$. $P=0.4196=42.0\%$.

4. Geometric $p=1/6$. $P=0.1157$. $E=6$.

5. $n>50, p<0.1$. $\lambda=8$. $P(X \leq 3)=0.0423$. Both $E=8$.